HW #2

Davis

Quarto

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Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

Introduction

This document analyzes a restaurant inspections dataset. The analysis includes data loading, exploration, cleaning, and visualizations to answer several questions related to inspection scores, trends over time, comparisons by city, inspector, facility type, and a focused analysis on restaurants.

Setup

We start by loading the required libraries.

```
library(readr)    # for reading CSV files
library(dplyr)    # for data manipulation
```

```

Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag


The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(ggplot2)  # for plotting
library(stringr)  # for string manipulation
```

Load the Data

Check the current working directory and load the dataset.

```
# Check current working directory
getwd()
```

```
[1] "/Users/SchoolAccount/Desktop/plan372_hm1repo/plan372_hmks/hw2"
```

```
# Load the dataset
restaruant_data <- read_csv("/Users/SchoolAccount/Desktop/plan372_hm1repo/HWS/restaurant_ins
```

```
Rows: 3875 Columns: 14
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (8): HSISID, DESCRIPTION, TYPE, INSPECTOR, NAME, RESTAURANTOPENDATE, CI...
dbl  (5): OBJECTID, SCORE, PERMITID, geocode_lon, geocode_lat
```

```
dttm (1): DATE_

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Data Exploration

Examine the dataset by checking its dimensions, the first few rows, summary statistics, column names, and overall structure.

```r
# Dimensions of the dataset: number of rows and columns
dim(restaruant_data)
```

```
[1] 3875   14
```

```r
# Display the first six rows of the dataset
head(restaruant_data)
```

```
# A tibble: 6 x 14
  OBJECTID HSISID SCORE DATE_               DESCRIPTION TYPE  INSPECTOR PERMITID
     <dbl> <chr>  <dbl> <dttm>              <chr>       <chr> <chr>        <dbl>
1 25137654 04092~  97   2017-10-22 04:00:00  <NA>       Insp~ Karla Cr~    13405
2 25115128 04092~  96   2019-02-27 05:00:00 "*Notice* ~ Insp~ Meghan S~    13939
3 25123164 04092~  98.5 2019-03-04 05:00:00 "*NOTICE* ~ Insp~ Kaitlyn ~    20554
4 25128895 04092~  90.5 2019-03-23 04:00:00 "Opening c~ Insp~ Angela M~    15506
5 25124786 04092~  97.5 2019-04-24 04:00:00 "*NOTICE* ~ Insp~ Patricia~    14839
6 25108274 04092~  98   2019-05-14 04:00:00 "*NOTICE* ~ Insp~ Maria Po~     8851
# i 6 more variables: NAME <chr>, RESTAURANTOPENDATE <chr>, CITY <chr>,
#   FACILITYTYPE <chr>, geocode_lon <dbl>, geocode_lat <dbl>
```

```r
# Summary of numeric columns
summary(restaruant_data)
```

```
    OBJECTID            HSISID              SCORE
 Min.   :25091064   Length:3875        Min.   :  0.00
 1st Qu.:25102773   Class :character   1st Qu.: 95.50
 Median :25118689   Mode  :character   Median : 97.50
 Mean   :25116443                      Mean   : 96.92
 3rd Qu.:25129320                      3rd Qu.: 98.50
```

```
 Max.   :25139899                      Max.   :100.00

     DATE_                         DESCRIPTION           TYPE
 Min.   :2017-10-22 04:00:00.00   Length:3875        Length:3875
 1st Qu.:2021-08-20 04:00:00.00   Class :character   Class :character
 Median :2021-10-28 04:00:00.00   Mode  :character   Mode  :character
 Mean   :2021-09-26 21:08:39.32
 3rd Qu.:2021-12-14 05:00:00.00
 Max.   :2022-01-31 05:00:00.00

  INSPECTOR           PERMITID         NAME           RESTAURANTOPENDATE
 Length:3875        Min.   :    1   Length:3875        Length:3875
 Class :character   1st Qu.: 6136   Class :character   Class :character
 Mode  :character   Median :12872   Mode  :character   Mode  :character
                    Mean   :12285
                    3rd Qu.:18374
                    Max.   :23602

     CITY           FACILITYTYPE        geocode_lon      geocode_lat
 Length:3875        Length:3875        Min.   :-78.94   Min.   : 0.00
 Class :character   Class :character   1st Qu.:-78.78   1st Qu.:35.75
 Mode  :character   Mode  :character   Median :-78.66   Median :35.80
                                       Mean   :-77.74   Mean   :35.37
                                       3rd Qu.:-78.60   3rd Qu.:35.85
                                       Max.   :  0.00   Max.   :36.05
                                       NA's   :296      NA's   :296
```

```r
# Names of the columns
names(restaruant_data)
```

```
 [1] "OBJECTID"           "HSISID"             "SCORE"
 [4] "DATE_"              "DESCRIPTION"        "TYPE"
 [7] "INSPECTOR"          "PERMITID"           "NAME"
[10] "RESTAURANTOPENDATE" "CITY"               "FACILITYTYPE"
[13] "geocode_lon"        "geocode_lat"
```

```r
# Structure of the dataset (includes types of each column)
str(restaruant_data)
```

```
spc_tbl_ [3,875 x 14] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ OBJECTID          : num [1:3875] 25137654 25115128 25123164 25128895 25124786 ...
```

```
 $ HSISID           : chr [1:3875] "04092021735" "04092013943" "04092021636" "04092021737"
 $ SCORE            : num [1:3875] 97 96 98.5 90.5 97.5 98 100 99.5 100 90.5 ...
 $ DATE_            : POSIXct[1:3875], format: "2017-10-22 04:00:00" "2019-02-27 05:00:00"
 $ DESCRIPTION      : chr [1:3875] NA "*Notice* Effective January 1, 2019, the NC Food Code
 $ TYPE             : chr [1:3875] "Inspection" "Inspection" "Inspection" "Inspection" ...
 $ INSPECTOR        : chr [1:3875] "Karla Crowder" "Meghan Scott" "Kaitlyn Yow" "Angela Mye
 $ PERMITID         : num [1:3875] 13405 13939 20554 15506 14839 ...
 $ NAME             : chr [1:3875] "THE CENTERLINE CAFE" "St. Raphael Hall Foodservice" "KN
 $ RESTAURANTOPENDATE: chr [1:3875] "2013/02/01 05:00:00+00" "2003/12/01 05:00:00+00" "2010/
 $ CITY             : chr [1:3875] "CARY" "RALEIGH" "KNIGHTDALE" "CARY" ...
 $ FACILITYTYPE     : chr [1:3875] "Food Stand" "Restaurant" "Food Stand" "Food Stand" ...
 $ geocode_lon      : num [1:3875] -78.8 -78.6 -78.5 -78.8 0 ...
 $ geocode_lat      : num [1:3875] 35.8 35.9 35.8 35.8 0 ...
 - attr(*, "spec")=
  .. cols(
  ..   OBJECTID = col_double(),
  ..   HSISID = col_character(),
  ..   SCORE = col_double(),
  ..   DATE_ = col_datetime(format = ""),
  ..   DESCRIPTION = col_character(),
  ..   TYPE = col_character(),
  ..   INSPECTOR = col_character(),
  ..   PERMITID = col_double(),
  ..   NAME = col_character(),
  ..   RESTAURANTOPENDATE = col_character(),
  ..   CITY = col_character(),
  ..   FACILITYTYPE = col_character(),
  ..   geocode_lon = col_double(),
  ..   geocode_lat = col_double()
  .. )
 - attr(*, "problems")=<externalptr>
```

Data Cleaning

Count missing values per column, filter out rows with missing values in vital columns (CITY, FACILITYTYPE, SCORE), and check for duplicate rows.

```
# Count missing data in each column
colSums(is.na(restaruant_data))
```

```
          OBJECTID             HSISID              SCORE              DATE_
```

```
                0                   0                   0                   0
      DESCRIPTION                TYPE           INSPECTOR            PERMITID
              516                   0                   0                   0
             NAME RESTAURANTOPENDATE                 CITY        FACILITYTYPE
              296                 296                 296                 296
      geocode_lon         geocode_lat
              296                 296
```

```r
# Filter out rows with missing values in CITY, FACILITYTYPE, and SCORE
restaruant_data <- restaruant_data %>%
  filter(!is.na(CITY) & !is.na(FACILITYTYPE) & !is.na(SCORE))

# Check the dimensions after filtering
dim(restaruant_data)
```

```
[1] 3579   14
```

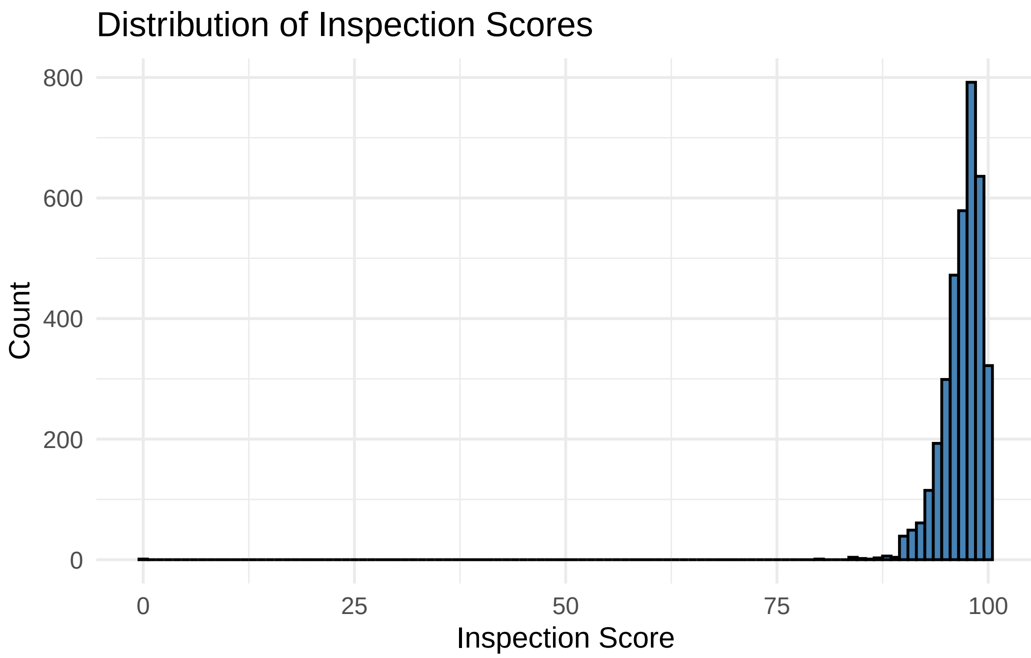
```r
# Check for any duplicate rows (should return 0 if there are none)
sum(duplicated(restaruant_data))
```

```
[1] 0
```

Question 1: Distribution of Inspection Scores

Visualize the overall distribution of inspection scores using a histogram.

```r
ggplot(restaruant_data, aes(x = SCORE)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Inspection Scores",
       x = "Inspection Score",
       y = "Count") +
  theme_minimal()
```



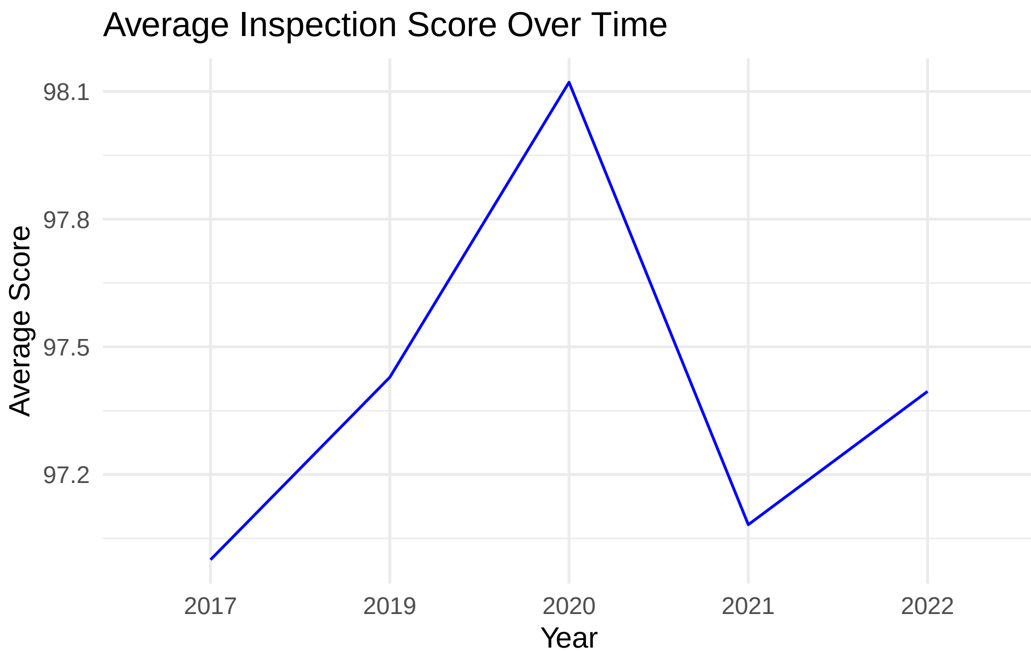
Question 2: Trends in Inspection Scores Over Time

Investigate whether there are trends over time by extracting the year from the `DATE_` column and calculating the average inspection score per year.

```
# Extract the year from the DATE_ column
restaruant_data$INSPECTION_YEAR <- format(restaruant_data$DATE_, "%Y")

# Group by year and calculate the average inspection score
avg_score_by_year <- restaruant_data %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE))

# Plot the trend over the years
ggplot(avg_score_by_year, aes(x = INSPECTION_YEAR, y = Average_Score, group = 1)) +
  geom_line(color = "blue") +
  labs(title = "Average Inspection Score Over Time",
       x = "Year",
       y = "Average Score") +
  theme_minimal()
```



Scores peaked in 2020 and were at their lowest in 2017.

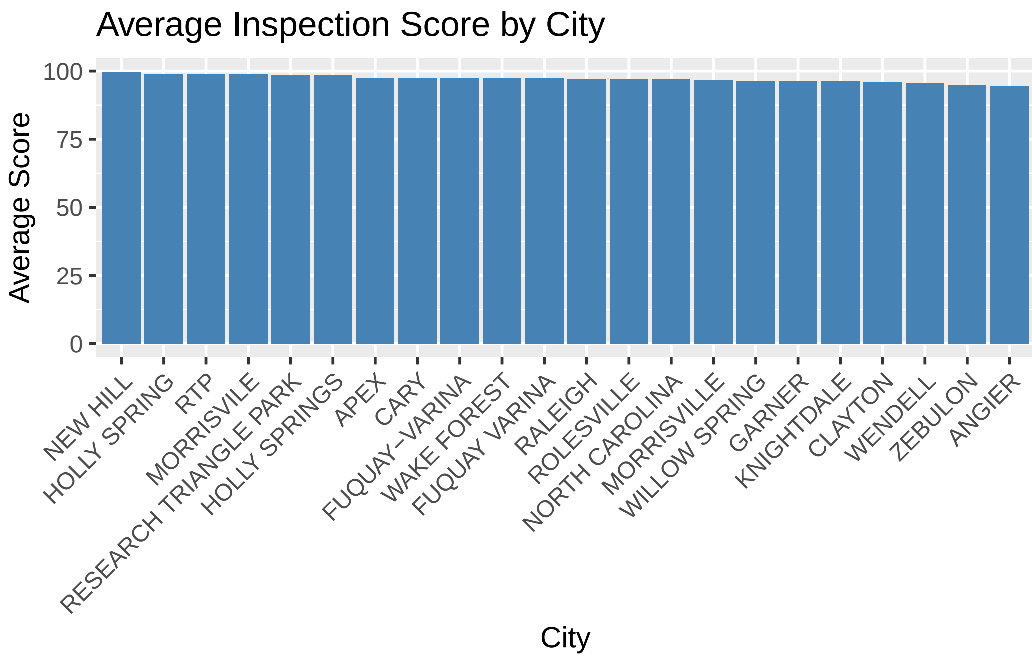
Question 3: Inspection Scores by City

Check if inspection scores vary by city by standardizing city names and calculating average scores.

```r
# Standardize CITY names to uppercase
restaurant_data$CITY <- str_to_upper(restaurant_data$CITY)

# Group by CITY and calculate average inspection score and count
avg_score_by_city <- restaurant_data %>%
  group_by(CITY) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE), Count = n())

# Create a bar plot for average scores by city
ggplot(avg_score_by_city, aes(x = reorder(CITY, -Average_Score), y = Average_Score)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Average Inspection Score by City",
       x = "City",
       y = "Average Score") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Inspection scores do not vary by city.

Question 4: Inspection Scores by Inspector

Summarize the average score and the number of inspections for each inspector.

```
inspector_summary <- restaurant_data %>%
  group_by(INSPECTOR) %>%
  summarize(
    Average_Score = mean(SCORE, na.rm = TRUE),
    Inspection_Count = n()
  ) %>%
  arrange(desc(Average_Score))

# Display the inspector summary table
inspector_summary
```

```
# A tibble: 39 x 3
   INSPECTOR           Average_Score Inspection_Count
   <chr>                       <dbl>            <int>
 1 James Smith                  99                  1
 2 Greta Welch                  98.5                2
```

```
 3 Kaitlyn Yow                98.5                 1
 4 Jason Dunn                 98.5                61
 5 John Wulffert              98.4               154
 6 Jamie Phelps               98.2               149
 7 Nicole Millard             98.1               146
 8 Zachary Carter             98.1               150
 9 Brittny Thomas             98                   3
10 Dipatrimarki Farkas        97.8               155
# i 29 more rows
```

Inspection scores basically do not vary by inspector.

Question 5: Sample Size Considerations

Examine the sample sizes for each group (city, inspector, and year) to assess if small sample sizes might affect the results.

```r
# Sample size by city
avg_score_by_city %>%
  select(CITY, Count) %>%
  arrange(Count)
```

```
# A tibble: 22 x 2
   CITY                    Count
   <chr>                   <int>
 1 ANGIER                      1
 2 HOLLY SPRING                1
 3 NORTH CAROLINA              1
 4 RESEARCH TRIANGLE PARK      1
 5 RTP                         1
 6 MORRISVILE                  2
 7 NEW HILL                    2
 8 WILLOW SPRING               2
 9 CLAYTON                     4
10 ROLESVILLE                 24
# i 12 more rows
```

```r
# Inspectors with less than 5 inspections
inspector_summary %>%
  filter(Inspection_Count < 5)
```

```
# A tibble: 5 x 3
  INSPECTOR      Average_Score Inspection_Count
  <chr>                  <dbl>            <int>
1 James Smith             99                  1
2 Greta Welch             98.5                2
3 Kaitlyn Yow             98.5                1
4 Brittny Thomas          98                  3
5 Thomas Jumalon          89                  3
```

```r
# Sample size by year
yearly_sample_size <- restaurant_data %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Inspection_Count = n()) %>%
  arrange(INSPECTION_YEAR)

# Display the sample size by year
yearly_sample_size
```

```
# A tibble: 5 x 2
  INSPECTION_YEAR Inspection_Count
  <chr>                      <int>
1 2017                           1
2 2019                          28
3 2020                          37
4 2021                        2853
5 2022                         660
```

There were a lot more inspections in certain cities than others (max 196, min 1). There were also a lot more inspections in the year 2021 (2853) than 2017 (1).

Question 6: Inspection Scores by Facility Type

Determine whether the scores for restaurants are higher than for other types of food-service facilities by grouping by `FACILITYTYPE`.

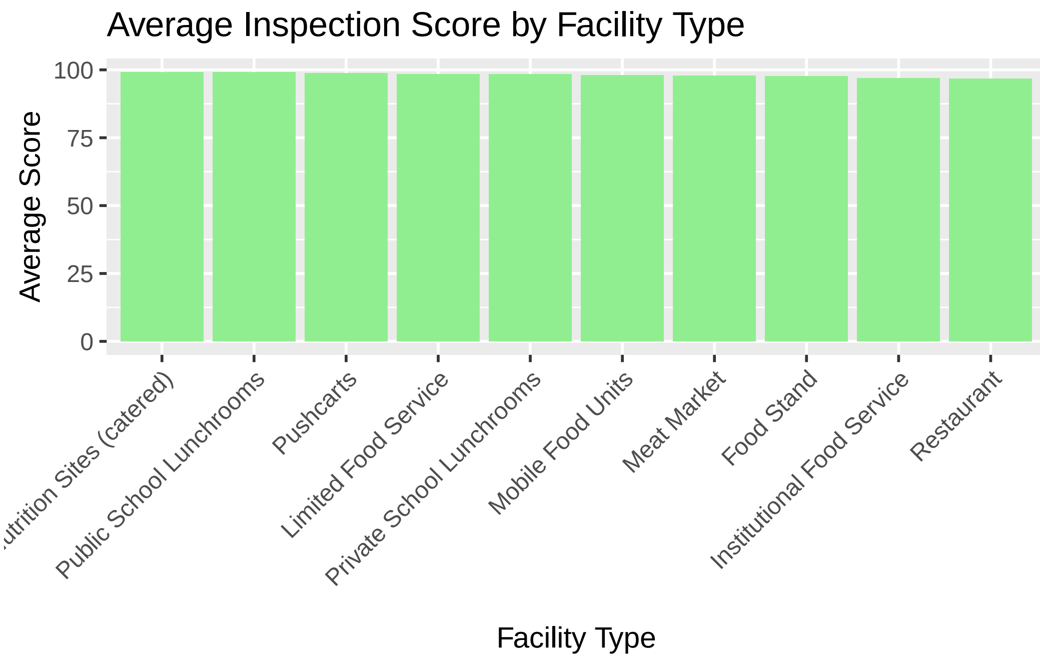
```r
# Group by FACILITYTYPE and calculate average score and count
facility_summary <- restaurant_data %>%
  group_by(FACILITYTYPE) %>%
  summarize(
    Average_Score = mean(SCORE, na.rm = TRUE),
```

```r
    Inspection_Count = n()
  ) %>%
  arrange(desc(Average_Score))

# View the facility summary table
facility_summary
```

```
# A tibble: 10 x 3
   FACILITYTYPE                      Average_Score Inspection_Count
   <chr>                                     <dbl>            <int>
 1 Elderly Nutrition Sites (catered)          99.2                8
 2 Public School Lunchrooms                   99.2              185
 3 Pushcarts                                  98.8               39
 4 Limited Food Service                       98.5                1
 5 Private School Lunchrooms                  98.5               13
 6 Mobile Food Units                          98.1              181
 7 Meat Market                                98.0               93
 8 Food Stand                                 97.7              661
 9 Institutional Food Service                 96.9               46
10 Restaurant                                 96.7             2352
```

```r
# Bar plot of average score by facility type
ggplot(facility_summary, aes(x = reorder(FACILITYTYPE, -Average_Score), y = Average_Score)) +
  geom_bar(stat = "identity", fill = "lightgreen") +
  labs(title = "Average Inspection Score by Facility Type",
       x = "Facility Type",
       y = "Average Score") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



All of the facilities are about the same. Nutrition sites and school lunchrooms generally score much better than restaurants.

Question 7: Analyses for Restaurants Specifically

Repeat the analyses above for restaurants only by first filtering the dataset.

```
# Filter the data for restaurants only
restaruant_only <- restaruant_data %>%
  filter(FACILITYTYPE == "Restaurant")

# Check the number of observations for restaurants
nrow(restaruant_only)
```

```
[1] 2352
```

7.1: Histogram of Inspection Scores for Restaurants

```r
ggplot(restaurant_only, aes(x = SCORE)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Inspection Scores for Restaurants",
       x = "Inspection Score",
       y = "Count") +
  theme_minimal()
```

7.2: Inspection Trends by Year for Restaurants

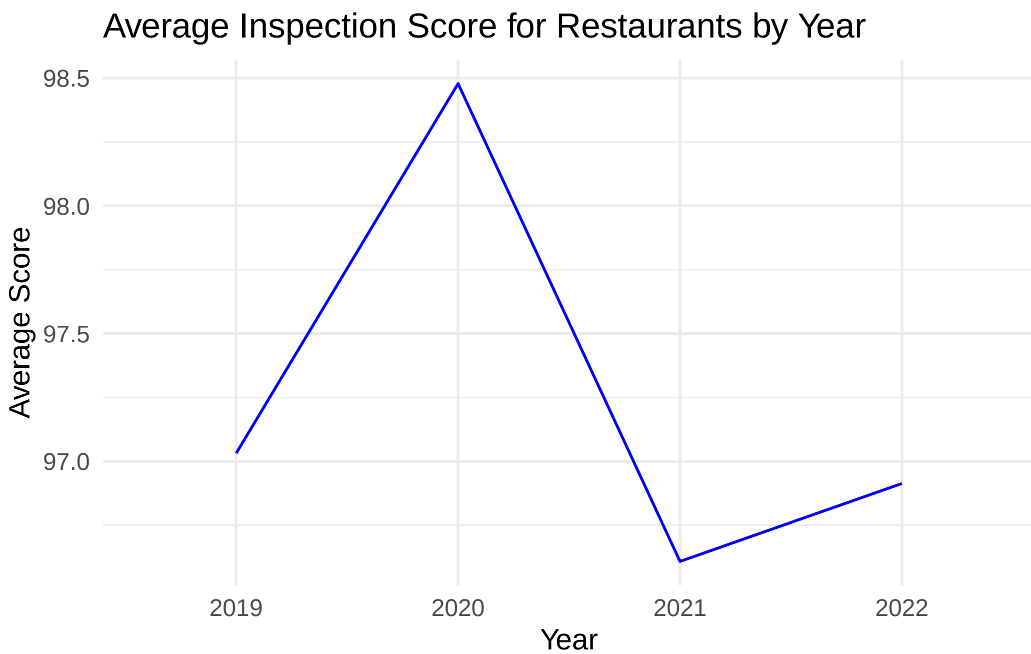
Extract the year for restaurants and plot the average inspection score by year.

```r
# Extract the year from the DATE_ column for restaurants
restaruant_only$INSPECTION_YEAR <- format(restaruant_only$DATE_, "%Y")

# Group by year and calculate the average score for restaurants
avg_score_restaurants_by_year <- restaruant_only %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE))

# Plot the trend over the years for restaurants
ggplot(avg_score_restaurants_by_year, aes(x = INSPECTION_YEAR, y = Average_Score, group = 1))
  geom_line(color = "blue") +
```

```
  labs(title = "Average Inspection Score for Restaurants by Year",
       x = "Year",
       y = "Average Score") +
  theme_minimal()
```



Peaked in 2020 and lowest in 2021.

7.3: Inspection Scores by City for Restaurants

```
# Group by CITY and calculate average score and count for restaurants
avg_score_restaurants_by_city <- restaruant_only %>%
  group_by(CITY) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE), Count = n())

# Display the table for restaurants by city
avg_score_restaurants_by_city
```

```
# A tibble: 21 x 3
   CITY          Average_Score Count
   <chr>                 <dbl> <int>
 1 ANGIER                 94.5     1
```

```
 2 APEX                      97.1   108
 3 CARY                      97.3   406
 4 CLAYTON                   93       1
 5 FUQUAY VARINA             96.9    49
 6 FUQUAY-VARINA             97.0    27
 7 GARNER                    95.8    93
 8 HOLLY SPRING              99       1
 9 HOLLY SPRINGS             98.0    79
10 KNIGHTDALE                95.1    49
# i 11 more rows
```

Lot more inspections in the cities with higher scores.

7.4: Inspection Scores by Inspector for Restaurants

```r
# Group by INSPECTOR and calculate average score and inspection count for restaurants
inspector_summary_restaurants <- restaurant_only %>%
  group_by(INSPECTOR) %>%
  summarize(
    Average_Score = mean(SCORE, na.rm = TRUE),
    Inspection_Count = n()
  ) %>%
  arrange(desc(Average_Score))

# Display the inspector summary table for restaurants
inspector_summary_restaurants
```

```
# A tibble: 38 x 3
   INSPECTOR           Average_Score Inspection_Count
   <chr>                       <dbl>            <int>
 1 James Smith                  99                  1
 2 Greta Welch                  98.5                2
 3 John Wulffert                98.1              111
 4 Brittny Thomas               98                  3
 5 Jamie Phelps                 97.9              108
 6 Zachary Carter               97.8              100
 7 Dipatrimarki Farkas          97.7              118
 8 Loc Nguyen                   97.6               74
 9 Ginger Johnson               97.6               35
10 Nicole Millard               97.6               48
# i 28 more rows
```

No major outliers, all about the same.

7.5: Sample Size Check for Restaurants

Examine sample sizes by city, inspector, and year for the restaurant-only subset.

```r
# Sample size by city for restaurants
avg_score_restaurants_by_city %>%
  select(CITY, Count) %>%
  arrange(Count)
```

```
# A tibble: 21 x 2
   CITY                    Count
   <chr>                   <int>
 1 ANGIER                      1
 2 CLAYTON                     1
 3 HOLLY SPRING                1
 4 MORRISVILE                  1
 5 NEW HILL                    1
 6 RESEARCH TRIANGLE PARK      1
 7 RTP                         1
 8 WILLOW SPRING               1
 9 ROLESVILLE                 13
10 WENDELL                    20
# i 11 more rows
```

```r
# Inspectors with fewer than 5 inspections among restaurants
inspector_summary_restaurants %>%
  filter(Inspection_Count < 5)
```

```
# A tibble: 5 x 3
  INSPECTOR      Average_Score Inspection_Count
  <chr>                  <dbl>            <int>
1 James Smith             99                  1
2 Greta Welch             98.5                2
3 Brittny Thomas          98                  3
4 Jason Dunn              97.1                4
5 Thomas Jumalon          88                  2
```

```
# Sample size by year for restaurants
yearly_sample_size_restaurants <- restaruant_only %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Inspection_Count = n()) %>%
  arrange(INSPECTION_YEAR)

# Display the sample size by year for restaurants
yearly_sample_size_restaurants
```

```
# A tibble: 4 x 2
  INSPECTION_YEAR Inspection_Count
  <chr>                      <int>
1 2019                          16
2 2020                          23
3 2021                        1916
4 2022                         397
```

Big differences in number of inspections in different cities and the different years.

Conclusion

Throughout this assignment, I relied on several key resources to guide my learning and help me complete the required analysis and visualizations. This experience not only reinforced my technical skills in R and Quarto but also deepened my understanding of how to use external resources effectively when working through complex problems. From the official documentation to community-based support on Stack Overflow, each resource played a distinct role in shaping my workflow and learning process.

The first and most essential resource I turned to was the official `ggplot2` documentation and cheat sheet. When it came to visualizing the inspection scores and trends over time, the `ggplot2` documentation was invaluable. It helped me figure out how to customize my plots and select appropriate aesthetics for the visualizations, such as using `geom_histogram()` for the distribution of inspection scores and `geom_line()` for showing trends over time. Initially, I struggled with formatting the x-axis labels for the city bar chart to avoid overlapping text. The documentation provided examples on how to rotate labels using `element_text(angle = 45)`, which solved that problem. These seemingly small adjustments improved the clarity and presentation of my visualizations significantly.

For data manipulation and cleaning, I leaned heavily on the `dplyr` documentation and examples from the R for Data Science book by Hadley Wickham. Specifically, functions like `group_by()`, `summarize()`, and `filter()` were central to creating summary tables for inspection scores by city, inspector, and facility type. While these functions are relatively straightforward, combining them for more complex operations required deeper understanding and trial and error. For example, calculating the average inspection score while ignoring missing values (`na.rm = TRUE`) initially returned incorrect results due to a filtering error. Referring back to the documentation helped me fix this issue, ensuring accurate calculations.

Another pivotal resource was the `kableExtra` package for generating polished tables. Creating summary tables that were visually appealing and easy to interpret was a priority for this assignment. I found several tutorials online that guided me through the process of applying conditional formatting to highlight key metrics in the table. These tables helped present the data more effectively, making it easier to draw comparisons between different cities and inspectors.

Stack Overflow was another critical tool in this learning journey. Whenever I encountered errors that weren't explained clearly in the R console, Stack Overflow was usually the first place I turned to. One of the most frustrating moments during this assignment occurred while trying to extract and format dates from the `DATE_` column for trend analysis. My initial approach led to an error that I couldn't understand. After searching the error message on Stack Overflow, I found a detailed explanation of how to use the `lubridate` package to simplify date handling. This not only fixed the problem but also introduced me to a new package that I hadn't used before, which I will definitely use again in future projects.